

ARender 2023.19.0

Reference benchmark — ~32,000 documents/hour sustained on a three-node Kubernetes infrastructure | Gatling, one-hour LOAD run

TEST DATE	VERSION	TOOL	DURATION	TEST SCOPE
April 27, 2026	2023.19.0	Gatling	1 hour	Rendition backend only

PREPARED BY

Petra — Uxopian Software

VERDICT: STABLE



On the reference infrastructure (three-node Linux Kubernetes cluster, 8 vCPU / 32 GB RAM per worker), the ARender 2023.19.0 Rendition backend sustains **~32,000 document opens per hour at 0% error rate** over 1,003,914 HTTP requests. Global latency: P50 17 ms, P95 131 ms, P99 269 ms. The benchmark covers the Rendition backend only — UI tier, document connector and DMS need to be sized separately in a real deployment.

1 Infrastructure & Test Scenario

REFERENCE INFRASTRUCTURE

Type	Production Linux Kubernetes cluster
Nodes	1 control plane + 3 workers
Server profile	8 vCPU, 32 GB RAM per worker node

TEST CONFIGURATION

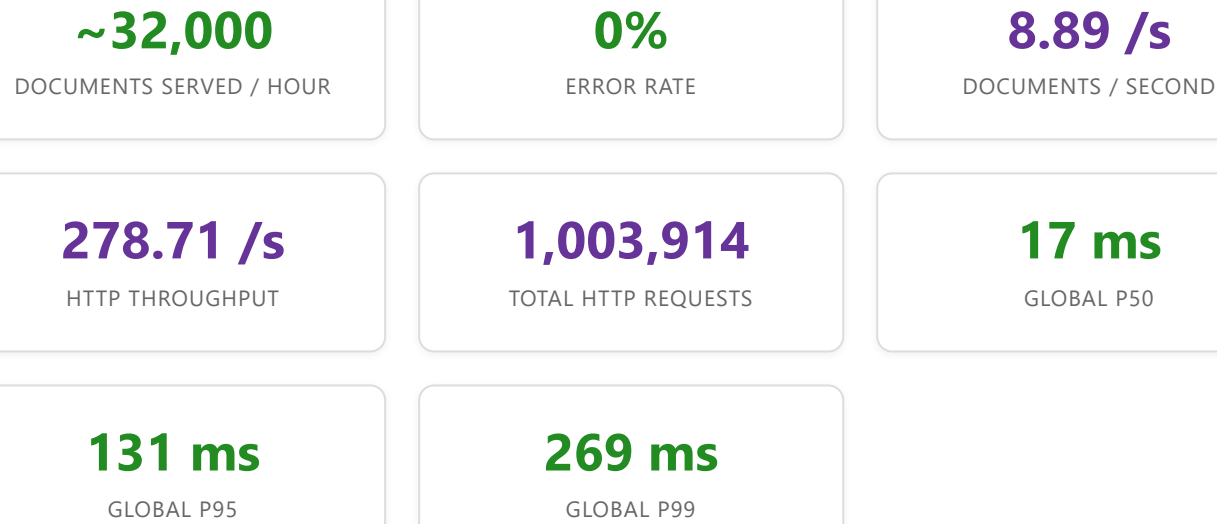
Tool	Gatling
Test type	LOAD
Duration	1 hour
Scope	ARender Rendition backend (broker + converter + renderer + text handler)

TEST SCENARIO — PER-DOCUMENT REQUEST SEQUENCE

- Step 1 Open_document_with_url
- Step 2 Get_document_layout
- Step 3 Get_3_images
- Step 4 Get_20_thumbnails
- Step 5 Get_3_pages_content
- Step 6 Evict_document

Injection mode: CLOSE. Each virtual user executes the full document lifecycle sequence from open to evict. The scenario simulates concurrent document viewing sessions.

2 Key Performance Indicators



3 Gatling Results — Standalone 3 nodes

Request	Total	OK	KO	% KO	Cnt/s	Min (ms)	P50 (ms)
All Requests	1,003,914	1,003,914	0	0	278.71	6	17
Open_document_with_url	32,039	32,039	0	0	8.89	14	18
Get_document_layout	32,039	32,039	0	0	8.89	78	228
Get_3_images	53,399	53,399	0	0	14.82	22	80
Get_20_thumbnails	224,279	224,279	0	0	62.27	7	25
Get_3_pages_content	630,119	630,119	0	0	174.94	6	15

Request	Total	OK	KO	% KO	Cnt/s	Min (ms)	P50 (ms)
Evict_document	32,039	32,039	0	0	8.89	6	10

4 Metric-by-Metric Analysis

Error Rate: 0% — All Requests Succeeded

All 1,003,914 requests returned successfully with zero KO. The system handled the full 1-hour load test without any functional failures, confirming strong reliability for ARender 2023.19.0.

P50 / P95 / P99 (All Requests): 17 ms / 131 ms / 269 ms

Median response time of 17 ms is excellent. P95 at 131 ms and P99 at 269 ms are well within acceptable ranges for a document viewing application under continuous load at ~279 req/s.

Get_3_pages_content: Highest Volume — Excellent Performance

This endpoint accounts for 62.8% of all requests (630,119 hits at 174.94 req/s). Response times are excellent: P50=15 ms, P95=41 ms, P99=62 ms, confirming that cached page content delivery scales well under load.

Get_20_thumbnails: High Volume — Good Performance

224,279 requests at 62.27 req/s with P50=25 ms, P95=54 ms, P99=73 ms. Thumbnail generation is well-optimized and stable.

Per-endpoint latency breakdown

Endpoint	P50 (ms)	P75 (ms)	P95 (ms)	P99 (ms)	Max (ms)	StdDev	As
Open_document_with_url	18	20	30	42	433	8	C
Get_document_layout	228	279	486	1,656	21,785	684	C
Get_3_images	80	193	240	273	924	77	C
Get_20_thumbnails	25	35	54	73	612	14	C
Get_3_pages_content	15	20	41	62	688	11	C
Evict_document	10	13	24	38	188	6	C

5 Resource Utilization

CPU

~30–40%

Node-level usage, stable across 3 workers. Viewer ~0.5 cores, Broker <0.5 cores, Converter <1 core, Handler <1 core, Renderer ~1 core per pod.

OK

MEMORY

~50–60% nodes

Node memory usage stable throughout the test. All components within configured limits.

OK

NETWORK

~150–400

Mbit/s

Network bandwidth ranged from ~100–300 Mbit/s in/out on nodes 1–2, up to ~400 Mbit/s peak on node 3. Stable throughout the test duration with no saturation.

OK

CPU: All Components Within Normal Range

All components (Viewer, Broker, Converter, Handler, Renderer) maintained stable CPU usage throughout the 1-hour test. Node-level CPU hovered between 25–40% across all 3 worker nodes with no saturation observed.

Memory: Stable

Node-level memory usage remained stable around 50–60% throughout the run, with all components staying within their configured limits.

Network: No Saturation

Network bandwidth remained well within capacity throughout the test. Peak observed was ~400 Mbit/s on the busiest worker node, with typical throughput around 150–250 Mbit/s. No packet loss or ingress saturation was detected.

6 Recommendations

1

Track `Get_document_layout` tail latency across releases

The maximum response time on this endpoint (21,785 ms) reflects an isolated outlier event during the run. Tracking the maximum and P99 across releases provides a useful regression indicator. An alerting threshold around P99 > 2,000 ms is a reasonable starting point.

2

Validate results with a longer stress test (2–4 hours)

The current 1-hour LOAD test confirms stability under steady-state conditions. A longer stress test would expose any slow resource trends and confirm the system remains stable across multiple document eviction cycles.

EXECUTIVE SUMMARY

On the reference infrastructure (three-node Linux Kubernetes cluster, 8 vCPU / 32 GB RAM per worker), the ARender 2023.19.0 Rendition backend sustains **~32,000 document opens per hour at 0% error rate** (1,003,914 HTTP requests, 278.71 req/s, 8.89 docs/s) on a one-hour run. Global latency: P50 17 ms, P95 131 ms, P99 269 ms. CPU, memory, and network resources are well within bounds for the duration of the run. **Verdict: STABLE — suitable for production at the tested load level.** The benchmark covers the Rendition backend only — UI tier, document connector and DMS need to be sized separately in a real deployment.